9.2. Neighborhood Q-learning

You might be wondering if there is a more efficient and compact way of representing actions and joint actions that might get around this issue of an impractically large joint-action space, but unfortunately there is no unambiguous way to represent an action using a more compact encoding. Try thinking of how you could communicate, unambiguously, which actions a group of agents took using a single number, and you'll realize you can't do it better than with an exponentially growing number.

At this point, MARL doesn't seem practical, but we can change that by making some approximations to this idealized joint-action Q function. One option is to recognize that in most environments, only agents in close proximity to each other will have any significant effect on each other. We don't necessarily need to model the joint actions of *all* the agents in the environment; we can approximate this by only modeling the joint actions of agents within the same *neighborhood*. In a sense, we divide the full joint-action space into a set of overlapping subspaces and only compute Q values for these much smaller subspaces. We might call this method *neighborhood Q-learning* or *subspace Q-learning* (figure 9.6).

Figure 9.6. In neighborhood MARL, each agent has a field of view (FOV) or neighborhood, and it can only see the actions of the other agents within this neighborhood. However, it may still get the full state information about the environment.